LAB 401 - Liver Function Tests: Full Lecture Map

Source-faithful one-page wall chart from: L1-LAB 401 - Liver function test

bilirubin | enzymes | synthetic function | jaundice patterns

1. Bilirubin Source + Metabolism

Aged RBC destruction
Reticuloendothelial system breaks hemoglobin into globin + haem.

Haem breakdown
Haem -> Fe + biliverdin -> bilirubin. Unconjugated bilirubin binds albumin.

Hepatocyte handling
Uptake by hepatocytes, conjugation, then secretion. A small part may regurgitate to blood.

Intestinal fate
Secreted into intestine -> stercobilinogen -> urobilinogen -> urobilin + stercobilin.

Reference range
Total bilirubin: 0.2-1.2 mg/dL Conjugated bilirubin: 0.0-0.2 mg/dL

Bilirubin disorder buckets
A. Unconjugated/indirect hyperbilirubinemia: 1. Overproduction: hemolysis, ineffective erythropoiesis. 2. Decreased uptake: Gilbert syndrome. 3. Decreased conjugation: neonatal jaundice, drug inhibition e.g. novobiocin, Crigler-Najjar.

B. Mixed conjugated + unconjugated: Hepatocellular diseases: acute hepatitis and chronic cirrhosis. C. Conjugated/direct: Impaired secretion: Dubin-Johnson, Rotor. Cholestasis: intrahepatic malignancy, primary biliary cirrhosis, drugs/oral contraception; extrahepatic gallstone or malignancy.

2. Main Liver Function Tests

Test	Reference / location	Clinical significance from lecture	Pattern / traps
ALT	< 40 IU/L; cytosol of hepatocyte	Aminotransferase; increased in all liver diseases.	Highest elevation 10-100 fold in hepatitis. More liver-centered than AST in this lecture map.
AST	< 45 IU/L; cytosol + mitochondria of hepatocyte	Aminotransferase; increased in all liver diseases.	2-5 fold in cirrhosis. Also present in muscles, kidney, RBCs.
ALP	50-130 IU/L adult; 2.5 fold in child; membrane-bound biliary tract enzyme	Cholestatic/biliary enzyme.	10 fold in extrahepatic cholestasis. 1-2 fold in hepatitis/cirrhosis. Also bone + placenta.
GGT	Up to 55 IU/L; membrane-bound biliary tract enzyme	Supports biliary tract/cholestasis pattern.	Rises with ALP in obstructive/cholestatic patterns.
PT	Functional test of coagulation factor synthesis	Liver is major site for coagulation factor synthesis.	Prolonged due to impaired synthesis in severe acute/chronic damage OR vitamin K deficiency in cholestasis.
Total protein	6-8 g/dL; albumin + globulins	Protein pool includes IgG, IgM, IgA fractions.	Chronic liver disease: increased globulins with decreased albumin synthesis.
Albumin	3.5-5 g/dL	Synthetic function marker.	Decreased synthesis in chronic liver disease.
Electrophoresis	Alpha 1, alpha 2, beta, gamma bands	Separates globulin fractions.	Lecture names fractions as part of chronic liver disease protein interpretation.

3. Enzyme + Protein Logic

Hepatitis
Aminotransferases show highest elevation: ALT/AST can rise 10-100 fold.

Cirrhosis
ALT/AST show lesser elevation: around 2-5 fold in the lecture.

Extrahepatic cholestasis
ALP/GGT show high elevation: about 10 fold.

Hepatitis/cirrhosis cholestatic enzymes
ALP/GGT show moderate elevation: about 1-2 fold.

PT prolongation
Due to impaired factor synthesis in severe acute/chronic liver damage or vitamin K deficiency in cholestasis.

Protein pattern
Chronic liver disease: decreased albumin synthesis + increased globulin fractions.

4. Jaundice Differentiation Table

Item	Hemolytic jaundice	Obstructive jaundice	Hepatic jaundice
Total bilirubin	Increased	Increased	Increased
Direct bilirubin	Normal	Increased	Increased
Indirect bilirubin	Increased	Slightly increased	Increased
ALT	Normal	Slightly increased	Increased
AST	Normal	Slightly increased	Increased
ALP	Normal	Increased	Slightly increased
GGT	Normal	Increased	Slightly increased
Prothrombin time	Normal	Prolonged	Prolonged
Urine bilirubin	Absent	Present	Present
Urine urobilinogen	Increased	Normal trace in incomplete obstruction; absent in complete obstruction	Slightly increased
Stool stercobilin	Increased	Pale stool in incomplete obstruction; clay-colored stool in complete obstruction	Slightly increased

5. Extra Lecture Enzymes

Amylase enzyme
Lecture headings to preserve: function, sources, P-amylase and S-amylase, clinical application in pancreatic diseases and salivary diseases.

Acid phosphatase
Tissue distribution: prostate, bone, RBCs. Lecture headings: fractions and clinical application in prostatic diseases, bone diseases, and hematologic diseases.

Note: amylase and acid phosphatase are included because the lecture explicitly lists them under clinical significance objectives.

6. Exam-Oriented Pattern Readout

Hepatocellular injury
ALT/AST dominate. Hepatitis gives 10-100 fold aminotransferase elevation.

Cholestasis/obstruction
ALP/GGT dominate. Extrahepatic cholestasis gives about 10 fold elevation.

Synthetic failure
PT prolongs when coagulation factor synthesis is impaired; albumin falls in chronic disease.

Bilirubin type first
Indirect points to overproduction/uptake/conjugation. Direct points to secretion/cholestasis. Mixed points to hepatocellular disease.

Lecture objectives preserved: recognize LFTs; differentiate hyperbilirubinemia causes; clinical significance of serum bilirubin, ALT, AST, ALP, GGT, amylase, acid phosphatase, total protein, albumin/protein electrophoresis, PT; select and interpret tests for diagnosis/follow-up of liver diseases.

Lecture Admin / Source Details
Clinical pathology / laboratory medicine. By Prof. Dr. Ahmed El Sayed Abd ALLAH El Tawil, Head of Clinical Pathology Department, Faculty of Medicine, MUST University. Pretest: 2 minutes on the new topic; What is clinical pathology? Is it important? Reference: McGraw Hill platform; Laposata's Laboratory Medicine: Diagnosis of Disease in the Clinical Laboratory, 3e, 2018; 6th year clinical pathology book, Faculty of Medicine Ain Shams University. LAB 401 marks: research 5, quiz/continuous assessment 10, final term 35.

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